Guided Flow Matching for Extreme Weather Simulation

Master's Thesis

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Abstract

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Chapter 1

Introduction

Machine learning models have reached state-of-the-art skill in weather forecasting, where they now rival traditional physics-based solvers at a fraction of their computational cost. One class among them are generative probabilistic models. First developed in the image domain, they have recently been carried over to weather. There, they now define the state of the art in probabilistic forecasting.

In the image domain, the applications of interest almost all *condition* a pretrained model rather than sample from it unconditionally: text-to-image generation, editing, and inpainting each steer the generative process towards a desired target through *guidance*. In the weather domain, by contrast, generative models are mostly used in their unguided form, as residual predictors that both overcome the oversmoothing of deterministic models and enable ensemble forecasts out of the box. The use of guidance in the weather domain remains largely unexplored. The few exceptions steer forecasts towards specific extreme events: [cite] generate windstorms, [cite] minimise precipitation, and [cite] produce heatwaves.

— frame it in the style of the presentation –> state what we want like in the presentation and also state the high level objective slide contents — we could also take inspiration from the proposal Extreme events cause a disproportionate share of weather-related damage, yet their rarity makes them precisely the cases in which forecast ensembles are thinnest and observational data for study scarcest. This motivates our central question: can we deliberately generate realistic extreme weather trajectories with a generative forecasting model, rather than wait for an unguided ensemble to sample them? Building on the recent works above, we tackle the question by splitting it into three subproblems—specifying the event, guiding the model, and evaluating the result—which together form a general framework to *specify*, *generate*, and *evaluate* extreme events with generative weather models.

But why do we need to adapt guidance methods from the image domain to the weather domain in the first place? The answer lies in three key differences between the two domains. — insert here the differences from the presentation

We build on ArchesWeather, a residual flow-matching model, and focus throughout on surface-temperature heatwaves over a chosen region of interest; the methodology itself is model and target agnostic. The three subproblems are also our three contributions:

1. **Specify.** A heuristic that uses historical states consistent with a given start state to define a plausible target trajectory for the region and variable of interest.
2. **Guide.** A gradient-based guidance method, adapted from classifier guidance, that steers the forecast onto the specified target by driving the gap to zero rather than by tuning guidance strength for a visually plausible result, thus requiring minimal tuning and minimising the intervention.
3. **Evaluate.** A set of flow and weather plausibility tests. In particular, for output plausibility we propose a benchmark, built from the same historical data, that tests generated trajectories for spatial, pressure-level, and cross-variable coherence, using the historical states as anchors and the unguided forecast as baseline.

The remainder of the thesis is organised as follows. **??** introduces the technical background on flow matching, guidance, generative weather models, and seminal applications of guidance in the weather domain. **??????** develop the three phases of the framework in turn: specifying the target event, designing the guidance, and defining the evaluation. **??** presents the experiments and **??** discusses them. **??** concludes with a summary, directions for future work, and final remarks.

Chapter 2

Background

2.1 Flow matching

Flow matching [2] models a data distribution p_1 as the endpoint of a continuous transport of a tractable prior $p_0 = \mathcal{N}(0, I)$. Let $(p_t)_{t\in[0,1]}$ be a *probability path* with these endpoints, generated by a time-dependent *velocity field* v_t through the flow

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\phi_t(x) = v_t\big(\phi_t(x)\big), \qquad \phi_0 = \mathrm{id}, \tag{2.1}$$

whose push-forward carries p_0 to p_t; equivalently, v_t and p_t obey the continuity equation $\partial_t p_t + \nabla\cdot(p_t\, v_t) = 0$. Sampling then reduces to drawing $x_0 \sim p_0$ and integrating the flow to $t = 1$.

Training regresses a network v_θ onto the field that generates p_t. This marginal field is intractable, but it decomposes as an expectation of *conditional* fields, $v_t(x) = \mathbb{E}_{x_1\sim p_{1|t}(\cdot\,|x)}\big[\, v_t(x \mid x_1)\,\big]$, each $v_t(\cdot \mid x_1)$ generating a simple path from p_0 to a fixed datum x_1. Flow matching exploits the fact that the tractable *conditional* objective

$$\mathcal{L}(\theta) = \mathbb{E}_{t,\, x_1\sim p_1,\, x_t\sim p_t(\cdot\,|x_1)}\Big\|v_\theta(x_t, t) - v_t(x_t \mid x_1)\Big\|^2 \tag{2.2}$$

shares its gradient with the intractable marginal one [2]. Under the linear (optimal-transport) path $x_t = (1-t)\,x_0 + t\,x_1$ with $x_0 \sim p_0$, the conditional field is the constant displacement $v_t(x_t \mid x_1) = x_1 - x_0$, so training amounts to least-squares regression onto $x_1 - x_0$.

At inference we draw $x_0 \sim \mathcal{N}(0, I)$ and integrate $\dot{x}_t = v_\theta(x_t, t)$ with a few Euler steps; independent draws yield independent samples. Since this map from x_0 to x_1 is a single deterministic and differentiable trajectory, its output can be steered by differentiating a target loss through the flow—the mechanism on which our guidance builds (**??**).

2.2 Guidance methods

Guidance methods were originally develped in the context of generative models based on diffusion. However, through the translation formulas from diffusion to flow matching we can easily identify their flow matching counterparts.

Classifier guidance

2.2.1 Backpropagating through the ODE

FlowGrad

D-Flow

Control paper

FlowGrad [3] backpropagates such a gradient through the sampling trajectory; D-Flow [1] instead optimises the source x_0, which for Gaussian paths projects the update onto the data manifold; and Wang et al. [4] frame the steering as optimal control. Our method adopts this gradient-based view but, rather than tuning a guidance strength, drives an explicit target gap to zero (**??**).

2.2.2 Classifier guidance

Classifier free guidance ...

We can translate the derived formula to flow land using following relation: ...

The general algorithm of classifier guidance works as follows.

...

A key finding is that in order to make guidance actually work, we have to abandon probability land, by increasing the strength of the guidance. In fact, when we start to play with *lambda*, the guided output is not a sample of the conditional probability distribution anymore.

2.2.3 Guidance schedules

The quest for the "best schedule" is still something that hasn't been solved in the guidance literature, although many heuristics have been proposed.

2.3 Generative weather models

Since the start of the machine learning weather literature a few models based on generative architectures have been proposed. Among the most prominent ones we can find GenCast.

GenCast ... operational ...

To prototype our method however the computational and memory requirements for GenCast are quite elevate. For this reason we pick as model of choice ArchesWeather.

2.4 ArchesWeather

We now spend some time presenting ArchesWeather in detail. This will enable us to introduce the relevant pieces we will be working on.

In our setting the model is a *residual* predictor: the generated x_1 is an increment added to a base state (a previous state or a deterministic forecast), so each ensemble member is that base plus one integrated trajectory (**??**).

2.4.1 Dataset

2.4.2 Architecture

2.4.3 Training

2.4.4 Evaluation

2.5 Guidance in weather modeling

The last piece of the puzzle before presenting our approach is to expose what people have alredy tried out, enabling us to expose the relevant gaps that we want to fill.

Digging into paper archives, I was able to find only three notable papers that were approaching the same general task as us.

The first one is ...

The most similar in nature, and the one from which we will blatently borrow some aspects of the evaluation, is also the one that approaches guidance and its components the most thorrouwly.

We have thus presented all relevant background pieces for the exposition of our contributions. We will spend the next three chapters introducing each of these key aspects.

2.6 Notation

Before delving into our method in the next chapters, we introduce here the notation we will use throughout the thesis.

Chapter 3

Specifying events of interest

3.1 Focus event of this project

In this thesis we explicitly set the focus on the heatwave event (TODO: state this in the intro directly). However, the method we develop is variable agnostic, and could be used and tuned to generate other extreme events of interest following a very similar logic.

We narrow the focus on purpose with the purpose of keeping the need for expert evaluation as far as possible from this project, in which we want to focus on the methodology itself.

3.2 Mask

In our setup we are interested in generating a heatwave scenario in a specific region of interest. This means, we want to guide the base forecast to generate higher temperatures over the selected region.

We define the region of interest as a simple bounding box, which defines a set of weigths (sum to 1) over a tensor having the same dimension as the weather states, non-zero at the variable and region of interest.

It is really important to understand how to evaluate what we are doing here. Since we are aiming at a specific region of interest, and we are guiding only wrt the region of interest, we are loosing the global coherence of the weather states. We shall thus evaluate the ability of our method to generate locally coherent weather scenarios, over the region of interest.

3.3 Target trajectory

In the following we define a climatological heuristic to define the target trajectory.

In order to define plausible average trajectories of change, we define following procedure: 1. Select start state 2. Define mask 3. Retrieve historical states that are consistent with the start state and the mask (+- some tolerance) 4. Compute the distribution of average trajectories of change for the historical states 5. Filter a quantile of the distribution to define the target trajectory (e.g. 0.95 quantile for extreme events) 6. Define a gaussian distribution at each time step of the target trajectory, with mean equal to the target trajectory and variance equal to the variance of the historical states at that time step. 7. Sample enough trajectories from the gaussian distribution to define a set of target trajectories that will be used to guide the generative model. 8. Select the target trajectories of interest in terms of extremeness to build scenarios. 9. Save the trajectories and corresponding states used to define the sampling distribution. These will be used to evaluate the plausibility of the generated trajectories as means of comparison. 10. Run benchmark tests and evaluate.

This heuristic enables us to anchor the guidance target and define a benchmark at the same time. More concretly, we avoid the issue of having to tune the strength of the guidance to achieve a certain plausible result. We reverse the problem by defining the plausible result first and then designing an algorithm to achieve that result and checking that both the algorithm and its result are well behaved.

We explicitly define a heatwave in order to design guidance methods. This means that we decide to represent a specific subset of state trajectoriesin this work. However, our method is flexible to any kind of target: you can swap your definition of the event you want to replicate and use the guidance to generate that kind of event. This really only affects the target definition.

This choice enables us to side step the problem that guiding towards a classifier's gradients or minimizing / maximizing a certain variable and stopping when we achieved the wanted result. If we specify the carachteristic of the wanted result we side step this problem entirely. The problem then shifts to retrieve a representative set of the event we have in mind. This is however a requirement for the first methodology. So we could say that the problem shifts towards defining an appropriate algorithm that can achieve a representative state similar to the specified set. Making this step explicit directly enables us to then compare the results against this set, and characterize its flaws / benefits directly.

Note that this whole pipeline enables us to sample trajectories wrt to the historical data. However, we can define arbitrary targets for our method, which we will do in chapter 6 when testing the method. In chapter 6.3 we will explore the usage of this heuristic detached from the method evaluation, in the context of the use cases we are actually interested in generating.

3.4 Pipeline summary

In summary we are explicitly defining our event of interest in terms of the average temperature over a specified area of interest, w.r.t. the climatology over the same area from the year 2020 to today.

— the concretization of our approach maybe go under experiments?

— maybe we don't need a summary section in the first place.

— put here a viz

Chapter 4

Guidance design

We start the chapter with the derivation of the general outline of the algorithm we will use for guidance. We then proceed in exposing different ways to specify its two main components: the loss and the strength-schedules. Importantly, the specific choices about the loss and the strength-schedule define the concrete algorithm variants that we derive from the general method.

4.1 Guidance base method

4.1.1 Classifier guidance

In our use case we want to generate weather states and guide them towards an extreme event of our choice (in our case increasing temperature). That is, we want to condition the generation on the property y, which encodes our preferences (target), in our case, the specification of extremeness level. In our context, unconditional generation is framed as sampling from the learned data distribution. The same way, the problem of guided generation can be framed as sampling from a from a posterior distribution.

Concretly, conditioning the residual latent z^t on a target property y gives

$$p(z^t \mid y) \propto p(y \mid z^t)\, p(z^t),$$
$$\nabla_{z^t} \log p(z^t \mid y) = \nabla_{z^t} \log p(z^t) + \nabla_{z^t} \log p(y \mid z^t).$$

4.1.2 Adapting classifier guidance

Originally the object $p(y \mid z^t)$ was computed using a classifier trained on the same set of images used to learn the data distribution. In our setting we define the target property through a loss $\mathcal{L}(\hat{x}_t)$ instead, which we make dependent on the clean prediction $\hat{x}_t$ at flow time t. Note, with this formulation we can compute the loss at any step of the flow on x_t with $0 < t < T$ and on the final clean prediction x_T. We define the likelihood through this loss, $p(y \mid x) \propto e^{-\mathcal{L}(x)}$ with $x := \hat{x}_T$ the clean state, and approximate the intractable expectation by the clean-state estimate at the current flow step t,

$$p(y \mid z^t) = \int p(y \mid x)\, p(x \mid z^t)\, dx = \mathbb{E}_{x|z^t}\left[e^{-\mathcal{L}(x)}\right] \approx e^{-\mathcal{L}(\hat{x}_t)}.$$

(this we do like in the DPS paper)

Substituting this approximation in the bayesian formula we get:

$$\begin{aligned}\nabla_{z^t} \log p(z^t \mid y) &= \nabla_{z^t} \log p(z^t) + \nabla_{z^t} \log e^{-\mathcal{L}(\hat{x}_t)} \\ &= \nabla_{z^t} \log p(z^t) - \nabla_{z^t} \mathcal{L}(\hat{x}_t)\end{aligned}$$

4.1.3 Guidance strength-schedule

Researchers in the image domain recognized that in order to obtain the aspired results, the guidance vector has to be scaled. The most naive way to accomplish this is to simply scale the guidance terms uniformly throughout the flow by multiplying by a constant w. Others apply a strength schedule λ_t. Others apply a normalization to step, such that the guidance vector represents a fix fraction of the unguided vector field.

The question for which guidance schedule is most optimal remains unanswered. In **??** develop variants of the base algorithm we will present next, testing empirically good choices for $\{\lambda\}_{t=1}^{T}$. Note that for $\lambda_t \neq 1$ for $t = 1, \ldots, T$, the exact posterior the guided results stops being a sample from the posterior distribution. In this sense we trade this mathematical reasoning for control.

— put this in the appendix or simply cite the MIT notes

In the context of flow matching, we can translate this form into:

$$\begin{aligned}\tilde{u}(z_t) &= u(z_t) - a_t \nabla_{z_t} \log p(y|z_t) \\ &= u(z_t) - a_t \nabla_{z^t} \mathcal{L}(\hat{x}_t)\end{aligned}$$

where $a_t = ...$ in the specific context of our model.

Applying the same reasoning about the guidance schedule in this context we rewrite the formula as

$$\begin{aligned}\tilde{u}(z_t) &= u(z_t) - a_t \nabla_{z_t} \log p(y|z_t) \\ &= u(z_t) - \lambda_t \nabla_{z^t} \mathcal{L}(\hat{x}_t)\end{aligned}$$

In this context, we can decompose $\lambda_t := w \cdot a_t \cdot c_t$. In our experiments we then use following nomenclature:

- λ_t: guidance schedule
- w: guidance strength (time independent)
- a_t: guidance profile, which doesn't have to be tied to the derivate a_t (TODO: use other form?)
- c_t: a step normalization constant, specified by the specific guidance algorithm

4.1.4 Base algorithm

Putting it all together we have described following base algorithm:

1. Estimate the clean state from the current noisy residual: $\hat{x}_t = \hat{x}^{\text{det}} + \hat{r}_t$.
2. Score it with the masked loss: $\mathcal{L}(\hat{x}_t)$.
3. Take the gradient with respect to the noisy state: $g_t = \nabla_{z^t} \mathcal{L}(\hat{x}_t)$.
4. Modify the velocity: $\tilde{u}_t = u_t - \lambda_t \, g_t$ with $\lambda_t = w \cdot a_t \cdot c_t$.
5. Take the Euler step as usual: $z^{t+1} = z^t + h_t \, \tilde{u}_t$.

Two design choices define the method:

- the **masked loss** — what to push and where;
- the **guidance schedule** — how hard and when.

Our implicit assumption here is that the gradient encodes the most likely shift to be applied under the learned data distribution to achieve the set target. This is only an hypothesis however and we do not test it. The reasoning we follow is the one adapted by the classifier guidance framework.

Note that, while classifier guidance based on a loss function is a viable option, researcher in the literature use the same base algorithm, but optimize the guidance term as adversarial noise instead of using the gradient. In our preliminary experiments we note that computing an adversarial perturbation to guide the generative model produces irrealistic results, compared to the gradient based perturbations. For this reason we use the adversarial perturbations as baseline.

We have thus defined previously the mask and the target profile in **??**. In the next we will address how these are used by the masked loss. We then address the question regarding the guidance schedule in the subsections exposing the three developed guidance algorithms.

4.2 Guidance loss

One of the main contributions of this thesis is the realization that in the weather domain, we can define a guidance target, which can be achieved precisely. By precisely I mean that the loss can be driven to zero. While in the image domain, there is always a ground truth during training, at inference we play with different guidance strengths to identify a range where the generated images are good looking. In the weather domain however, we can define a target that can be closed by tuning the guidance strength.

This is great because instead of tuning the guidance strength to find a range of plausible results, we can define what plausible results look like a priori (guidance target sampling trajectory defined through historical data and plausibility tests), and optimize the guidance strength to achieve this target.

Remember from section 3 that we decided to prescribe the guidance target value at each weather step.

We define the guidance loss as follows:

$$\begin{aligned} \mathcal{L} &= \Big(\underbrace{\mathcal{M}(\hat{x}_t)}_{\text{current}} - y \Big)^2 = \xi_t^2 \\ \mathcal{M}(x) &= \sum m \odot x \,, \quad \text{masked average with } \sum m = 1 \end{aligned}$$

where $\hat{x}_t$ is the clean-state estimate at flow step t and e_t is the gap (signed difference).

We realize that since we are pushing the base forecast (gui–ung object), this is equivalent to state the following:

$$\mathcal{L} \;=\; \Big(\underbrace{\mathcal{M}(\hat{x}_t)}_{\text{current}} \;-\; \underbrace{(1+\phi_n)\,\mathcal{M}(x^{\text{gui\_ung}})}_{\text{target}} \Big)^2$$

ϕ_n *section*

where ϕ_n is the intervention profile (the imposed change) in percetage. The reason to do this is that it provides a practical way to visualize guided results against the base forecast from different perspectives, which also include ϕ. Another one is that this enables us to test our methods against intensity levels in terms of simple percentages, without the need to go through the pipeline defined in the previous chapter.

In addition to the gap ξ, we could to the loss arbitrary regularization terms encoding different desiderata. Writing it down explicitly, the loss would then take the general form:

$$\mathcal{L} \;=\; \xi^2 + \gamma\mathcal{R}$$

Note that the gradient flows through the model into the full noisy residual z_t, because the network Jacobian is defined over the whole state.

We essentially have a very similar setting to the guided diffusion for precipitation paper, but we make it more general in these ways:

- we define a guidance target explictly, in contrast to observe results across different lambdas
- we experiment with different a_t trajectories, in addition to guiding computing the gradient at the first step
- we test level of intensity by shifting the intenisty parameter from the flow (lambda) to the trajectory definition (phi_n)
- Our loss is driven to zero in terms of the difference from the target, instead of being minimized iteratively (a more explicit way to encode the target scenario)

4.3 Guidance schedule (specific algorithms)

In this section we develop different variants of classifier free guidance, which we adapt to this setting. We explore the different variants, tune them, expose their flaws, assess their computational requirements, and compare their respective trade off. In the end we close in on the most promising method in terms of reliability and computational requirements, evaluated both in terms of theoretical cost, but also in terms of practical factors such as difficulty of tunining and potential need for multiple runs.

4.3.1 CLOSE-GAP

Given the loss defined above, we can define a guidance method that simply closes the gap to the target as prescribed by the guidance schedule in a greedy fashion.

At each step, the algorithm performs following tasks:

- Compute the gap and the gradient
- Compute the guidance factor $lambda_t$ s.t. $m(step + 1) - m(step) = (1 - \eta) \cdot e$ (by normalizing the gradient)

At the end of the flow we will have closed the gap up to $(1 - n)^{T-1}$.

As we will present in the experiments section, the method tends to produce a high oscillatory behavior, because the unguided steps will overshoot the scheduled gap if we push too much. This happens whether we want to close the theoritical gap or the actual gap.

If we knew a priori the optimal guidance strength to apply at each step, the CLOSE-GAP method would be optimal in terms of convergence and stability. However, we do not know the optimal guidance strength, and therefore we have to tune it. This means that we trade off the number of optimization steps we take to find the optimal guidance strength, with the number of experiments to tune for the optimal guidance strength schedule.

The method thus trades its efficient greedyness, with the need to tune η, before optaining a more well behaved guided trajectory, thus requiring multiple runs, making it much more costlier in its practical application.

4.3.2 OPTIMIZE-SCHEDULE

We try to overcome the issue of the fix step sizes, by trying to optimize λ. We can encode the well-behaviourness criteria directly into the loss function, by penalizing the total guidance needed to reach the set target. Ideally this would not generate the iscillatory behaviour, since then we would need to guide downwards. We could dream up arbitrary penalty terms such as angular defelction, ... but we keep it simple and start experimenting with this term here.

We add the regularization term to the loss as follows: ...

We then optimize lambda_t by using a ... method.

Note that this time the step is allowed to take the gradient norm into consideration.

In theory this method promises to find us the schedule that minimizes the total guidance needed to obtain the guidance target. However in practice this optimization is really hard to perform, because of the high dependance of λ^* on its initialization. The problem is related to the fact that the gradient applyed for the guidance at each step changes with respect to the schedule. So the optimizer will converge very fast to a local minimum, highly influenced by the initialization of the schedule.

4.3.3 OPTIMIZE-GAIN

As final method we develp one that solves the oscillatory behaviour of the first method, and removes the high dependency on the initialization of the second method, by encoding it explicitly, thus requiring lower computational requirements.

The approach of this method is to fix the a_t schedule (in contrast to the second method), but allowing the gradient magnitude to play a role (in contrast to the first method), and optimize for a w that achieves the set target over the T flow steps.

We realize that the dependence on w of the loss is almost linear (TODO: check empirically), so we can use a simple secant method that converges realiably and fast.

More concretly, the algorithm works as follows:

- Initialize w to a lower value to one that we know works empirically
- Make the flow with that w
- Compute the loss and compute two secant points
- Set the middle point as the new w

Across the methods this is the one that is easier to tune and works most reliably across settings. We will see this empirically in section 6.

4.3.4 Comparison

TABLE

- approach
- hyperparameters
- time complexity

- memory requirements

Chapter 5

Evaluation

We evaluate our guidance method against the two requirements we set for it: *achievement*, the ability to reach the specified target, and *realism*, the physical plausibility of the states it generates.

The first requirement concerns whether the algorithm reaches the target, and how well it does so across settings. We test this through convergence, stability, and trajectory deviation, which together expose the failure modes of the algorithm and reveal which design choices are more adequate overall.

The second requirement concerns whether the generated states are realistic, in terms of temporal, spatial, vertical, and cross-variable coherence.

The evaluation is organised as a funnel. In *flow-time* we assess the sampling dynamics of the algorithm variants and guidance schedules, and select the most promising method. In *weather-time* we then characterise the parameters of the experimental setting, select the settings that produce the most desirable results, and finally validate the realism of the generated trajectories. **??** carries out each stage in turn and reports the concrete settings and results; this chapter defines only *what* we test and *how* we score it.

Alongside these tests, the discussion examines further aspects that the framework does not measure directly—such as the diversity of the generated trajectories and the informativeness of the gradient—but that bear on the overall value of the method and its applications.

5.1 Flow-time assessment

Here we assess the algorithm variants and the guidance schedules, with the purpose of selecting the most promising method and parameters for the next set of experiments. Before asking whether the method produces realistic results, we first have to establish that it works: that it reaches the target, does so reliably across settings, and does so in a stable manner.

5.1.1 Tests

We test the achievement requirement and well–behaveheadness of the trajectory for different guidance algorithms through the following tests:

T1 – Convergence
The gap to the target is driven to zero. We report the mean and standard deviation of the final gap.

T2 – Stability
The trajectory is free of oscillatory behaviour. We measure this simply with respect to the pushback measured in the average trajectory (gui vs. ung steps).

T3 – Trajectory deviation
The guided trajectory does not deviate excessively from the unguided one. We report the distance between the two in PCA latent space, and the cosine similarity between the steps.

T4 – Tuning complexity
The algorithm parameters can be tuned reliably such that it combines its theoretical desirable properties to its practical applicability (TODO: should be the name of the test).

It's not clear a priori whether total guidance needed to achieve the target is also something to aim for. For completeness we add this in the stability test reports.

We use these tests as criteria to select the most promising method and its an adequate set of parameters for the tests at weather time.

5.2 Weather-time assessment

In weather-time we first characterise the parameters of the experimental setting—to describe their properties and to select settings for the larger experiments—and then validate the realism of the resulting trajectories. Throughout, we pay special attention to the diversity of the produced results as a secondary property of interest, alongside realism.

5.2.1 Parameter characterisation tests

In the following we provide a set of qualitative tests to better understand the role of the main parameters of the guidance at weather time. We test parameters in isolation, using the remaining parameters as controls. We proceed in three stages, examining in turn the region of interest, the variable of interest, and the intervention profile. For each we show different charts and metrics to expose what happens across these settings.

— maybe here is where the baseline would shine.

The reason for these tests is mostly to enhance our intuition about the role of each parameter, and provide a guideline on to select parameters for the large use cases we aim to produce with this work.

Region of interest

Here we assess variability of results by changing size, shift (from center), and coast-land covering of region of the interest.

Variable of interest

For simplicity we focus on heatwaves, and therefore on surface temperature. To probe the informativeness of the gradient, in this test we push the most correlated variables in an attempt to reproduce the pattern obtained by pushing temperature directly.

Intervention profile

Here we assess the degradation of the target patterns across intensity profiles. The intuition is that at some point things should brake down in some trivial way.

5.2.2 Realism tests

Once the guidance method and its parameters are selected, we assess the realism of the generated trajectories. Realism is central to this work, because we deliberately guide the generative model, possibly outside its training distribution. The tests themselves, however, are not specific to this work: they concern the general problem of evaluating the coherence of data-driven models, namely that the generated states should be consistent with the ground-truth dataset in both the guided and unguided settings.

We assess coherence to the ground-truth dataset along four dimensions:

T4 (Temporal).
: Transitions of a variable across weather steps should be smooth.

T5 (Spatial).
: Patterns over the region of interest should be spatially coherent.

T6 (Vertical).
: Patterns should be coherent across pressure levels. We test this using a level rank coherence metric.

T7 (Cross-variable).
: Patterns should be coherent with the learned weather transition function across variables. However, this is difficult to test quantitatively, we thus only make a qualitative assessment. (TODO: use the expert knowledge here)

We benchmark these properties against the ground-truth data as anchor (reference set), and against the unguided forecast as baseline, comparing unguided, guided-minus-unguided, and guided states. For T4–T6 we compute difference maps along the axis of interest and report the min, mean, max, and standard deviation across maps. For T7 we compute the cross-variable distributions over the weather steps of the ground-truth data, and report the deviation of the unguided, guided-minus-unguided, and guided states from their average.

For all four tests we average over the experiments and report the mean and standard deviation across runs. This indicates whether the generated extremes are faithful to the data distribution. A subtle but important point: for this to be meaningful, the base forecast must have missed the potential heat spots. The extreme is not only about the average—which the forecast most likely covers faithfully—but about the extreme spots, so both the mean and the max must be tested.

?? sets out the concrete settings for each stage of this framework and reports the results, which **??** then discusses.

Chapter 6

Experiments

This chapter carries out the evaluation framework of ??. Each section follows the same pattern: we first state the concrete settings, then report the results and the decision they lead to. Tests are referenced by the identifiers introduced in ?? (T1–T7).

For all experiments we select the region of interest and retrieve the state corresponding at the start of the heatwave reaching the highest temperatures over the mask with selection criteria being the highest rolling window sum over 5 days.

6.1 Flow-time evaluation

6.1.1 Setup

Here we assess the performance of the three proposed guidance algorithms with following setup:

- 3 different starting states with different masks (Europe, America, Australia)
- N=2, M=3
- 2 intensity levels (+0.25%, +0.75%)
- 2 a_t schedules (spike, closegap($\eta = 1$))

6.1.2 Achievement tests

The difficulty of tuning the first and second algorithm make them less suitable than the third one. Regarding algorithm 1, having a lower computational complexity is useless in practice if we then have to perform multiple costly runs. Regarding algorithm 2, having a desirability requirement encoded in the loss itself only complicate things. Tuning the optimization of this algorithms outweights the benefits, and induces a much higher cost in practical terms to the algorithm.

At later stages, the gradient becomes more similar to the mask (is more diluted, less sparse), which makes the guided pattern at the end of the flow be more spread.

Using a spike@1 approach is essentially equivalent to shifting the starting point of the flow trajectory (especially for the region over the mask) and let particle flow across the learned ODE.

What we observe is that the model doesn't necessarily pushes back against us. In contrary, when the method is tuned well, the model will do most of the pushing itself (look at this in percentage).

6.2 Weather-time evaluation

Throughout the weather-time experiments we vary the start state while fixing $N = 2$ and $M = 3$: this controls for the effects of $N = 1$ and for seed-specific effects, while keeping computational and memory requirements as low as possible.

Note that we will use the same experiments for the parameter characterization tests and for the realism tests.

— the realism tests could simply be used in each parameter subsection

6.2.1 Parameter characterisation tests

Region of interest

Variable of interest

Intervention profile

6.2.2 Realism tests

6.3 Use cases

For the final use cases we fix the parameters selected above and increase N and M, producing the experiments we had in mind when setting the goals of the project.

6.3.1 Europe 2020-06-01

6.3.2 Australia 2020-01-01

6.3.3 North America 2020-07-01

Chapter 7

Discussion

7.1 Mask

We experimented with a gaussian mask, but realized in our experiments that the best thing to do is equal weighting inside the region. The correct way to define the region of interest is to choose a greater mask size instead of using some weighting scheme fading out of an epicenter. The mask size should be chosen to cover the entire area of interest, ensuring that all relevant data is included in the analysis. A larger mask size can help to reduce noise and improve the accuracy of the results, while a smaller mask size may exclude important information. It is important to carefully consider the appropriate mask size for each specific application to achieve optimal results. Either way we (most probably) loose the global coherence for the guided weather states, so we might as well define directly the region of interest with a mask. Additionally, this enables us to actually make statements about "the region of interest", having equal weights everywhere. Introducing different weigths might bias the generation in a way that is not desired.

We will see that including or not a specific area will change the guided patterns substantially. This however is consistent with the notion of "region of interest". If we consider a greater area for the changes, the changes will be applied respectively.

7.2 ...

The algorithm doesn't produce diverse results, because the gradients are really stable. This has to do with the fact that we have learned one and only one flow model (transition function). If we had multiple residual diffusion model, we would expect the possible effects to be more diverse. However, the second interpretation might be that the gradient provides with the information about the most likely direction of change (which is true under the learned data distribution).

This happens because the guidance vector is a scaled version of the flow model's gradient wrt to the noisy state. The gradient is stable across the full trajectory, because the only variable inputs are the noise itself and the timestep of the flow. The stochasticity of the noise is not enough to produce diverse results. The direction of the gradient is always the same. If we say "let the masked average go up by phi percent", the only thing that changes is the magnitude of the gradient, but not its direction, which is specified by the Jacobian.

This is in contrast to guidance settings.

Shift between atmosphere and stratosphere.

Put here all observations

Chapter 8

Conclusion

8.1 Summary

8.2 Future Directions

8.3 Conclusive Remarks

Appendices

Appendix A

Code-base structure

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